

1. (4 Points)

Find the first two nonzero terms of Maclaurin series for $\cos^2 x$.

- ☐ $1 - \frac{x}{2!}$
- ☐ $1 - \frac{x^2}{2!}$
- ☐ $x - \frac{x^3}{3}$
- ☒ $1 - x^2$ ✓

2. (4 Points)

Find a Cartesian equation for the curve represented by the given Polar equation $r = \frac{1}{2 - 2 \sin \theta}$.

- ☒ $4x^2 - 4y = 1$ ✓
- ☐ $x - 4y^2 = 1$
- ☐ $4x^2 - 4y^2 = 1$
- ☐ $-x^2 - y^2 = 0$

3. (4 Points)

Find $\lim_{x \rightarrow \infty} \frac{x^{50}}{e^x}$.

- ☒ 0 ✓
- ☐ e^{50}
- ☐ ∞
- ☐ e^{-50}

4. Find the points on the curve (4 Points)

$x = 1 - 2 \cos t$, $y = 1 - 2 \sin t$, $0 \leq t \leq 2\pi$, where the tangent line is horizontal.

- ☐ (1, 0) and (1, 2)
- ☐ (1, 1) and (-1, 3)
- ☒ (1, -1) and (1, 3) ✓
- ☐ (1, 1) and (1, 2)

5. (4 Points)

Find the area of the region enclosed by the rose $r = 2 \sin 2\theta$.

- ☐ $\frac{\pi}{2}$
- ☐ π
- ☐ $\frac{3\pi}{2}$
- ☒ 2π ✓

6. (4 Points)

Find the length of the hyperbolic function $y = \frac{1}{4} \cosh 4x$, $0 \leq x \leq 2$.

- ☐ $\frac{\sinh(4)}{2}$
- ☒ $\frac{\sinh(8)}{4}$ ✓
- ☐ $\sinh(4)$
- ☐ $\sinh(8)$

7. Which of the following represents the area of the surface obtained by rotating the given curve about the x-axis (4 Points)

$x(t) = t - \sin(t)$, $y(t) = 1 - \cos(t)$, $0 \leq t \leq \frac{\pi}{2}$.

- ☐ $2\pi \int_0^{\frac{\pi}{2}} (1 - \cos t)^{1/2} dt$
- ☐ $2\sqrt{2}\pi \int_0^{\frac{\pi}{2}} (1 - \cos t)^{1/2} dt$
- ☐ $\pi \int_0^{\frac{\pi}{2}} (1 - \cos t)^{3/2} dt$
- ☒ $2\sqrt{2}\pi \int_0^{\frac{\pi}{2}} (1 - \cos t)^{3/2} dt$ ✓

8. (4 Points)

Which of the following series is convergent

- ☐ $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n^2}}$
- ☐ $\sum_{n=1}^{\infty} \cos\left(\frac{5}{n}\right)$
- ☐ $\sum_{n=0}^{\infty} \left(-\frac{5}{4}\right)^n$
- ☒ $\sum_{n=1}^{\infty} \frac{n^2 + 5}{(n + 2)!}$ ✓

9. Determine whether the following improper integral is convergent or divergent. If it is convergent evaluate it. (4 Points)

$\int_0^{\infty} x^3 e^{-2x^4} dx$.

- ☐ Convergent; $\frac{1}{4}$.
- ☒ Convergent; $\frac{1}{8}$. ✓
- ☐ Convergent; $\frac{1}{2}$.
- ☐ Divergent.

10. The radius of convergence of the series (4 Points)

$\sum_{n=1}^{\infty} \frac{n^{2n}}{(2n)!} (x - 1)^n$ is equal to

- ☐ $\frac{4}{e}$
- ☐ $\frac{2}{e}$
- ☒ $\frac{4}{e^2}$ ✓
- ☐ $\frac{2}{e^2}$

11. Find the area and the x-coordinate of the centroid of the region bounded by the curves (4 Points)

$y = x^2$ and $y = 2x - x^2$.

- ☒ $A = \frac{1}{3}$; $\bar{x} = \frac{1}{2}$ ✓
- ☐ $A = \frac{1}{6}$; $\bar{x} = \frac{2}{5}$
- ☐ $A = \frac{1}{2}$; $\bar{x} = \frac{1}{5}$
- ☐ $A = \frac{1}{3}$; $\bar{x} = \frac{1}{6}$

12. (4 Points)

A series expansion of $\frac{x - \sin x}{x^2}$ is

- ☐ $\frac{1}{2!} - \frac{x^2}{4!} + \frac{x^6}{6!} + \dots + \frac{(-1)^{n+1} x^{2n-2}}{(2n)!} + \dots$
- ☐ $\frac{x}{2!} - \frac{x^3}{4!} + \frac{x^5}{6!} + \dots + \frac{(-1)^{n+1} x^{2n+1}}{(2n)!} + \dots$
- ☒ $\frac{x}{3!} - \frac{x^3}{5!} + \frac{x^5}{7!} + \dots + \frac{(-1)^{n+1} x^{2n-1}}{(2n+1)!} + \dots$ ✓
- ☐ $\frac{x^2}{3!} - \frac{x^4}{5!} + \frac{x^6}{7!} + \dots + \frac{(-1)^{n+1} x^{2n}}{(2n+1)!} + \dots$

13. Find the area of the surface generated by rotating the given curve about the x-axis (4 Points)

$y = \sqrt{2 - x^2}$, $-1 \leq x \leq 1$.

- ☒ $4\sqrt{2}\pi$ ✓
- ☐ $8\sqrt{2}\pi$
- ☐ 4π
- ☐ 8π

14. What is the correct form of the partial fraction decomposition? (4 Points)

- $\frac{2x + 1}{(x + 2)^2(x^2 + 6)^3}$.
- ☐ $\frac{A}{x + 2} + \frac{B}{(x + 2)^2} + \frac{Cx + D}{(x^2 + 6)^3}$
- ☐ $\frac{A}{x + 2} + \frac{B}{(x + 2)^2} + \frac{C}{(x^2 + 6)} + \frac{D}{(x^2 + 6)^2} + \frac{E}{(x^2 + 6)^3}$
- ☐ $\frac{A}{(x + 2)^2} + \frac{Bx + C}{(x^2 + 6)} + \frac{Dx + E}{(x^2 + 6)^2} + \frac{Fx + G}{(x^2 + 6)^3}$
- ☒ $\frac{A}{x + 2} + \frac{B}{(x + 2)^2} + \frac{Cx + D}{(x^2 + 6)} + \frac{Ex + F}{(x^2 + 6)^2} + \frac{Gx + H}{(x^2 + 6)^3}$ ✓

15. (4 Points)

Find the appropriate trigonometric substitution for the integral $\int \frac{x^3}{(3 + 4x - 4x^2)^{3/2}} dx$.

- ☒ $x = \frac{1}{2} + \sin \theta$ ✓
- ☐ $x = \frac{1}{2} + \tan \theta$
- ☐ $x = \frac{1}{2} + \sec \theta$
- ☐ $x = 2 + \tan \theta$

16. Determine if the following series converges or diverges. If the series converges, find its sum (4 Points)

$\sum_{n=1}^{\infty} 7^{-n} 2^{n+1}$.

- ☐ Converges; $\frac{1}{5}$
- ☒ Converges; $\frac{4}{5}$ ✓
- ☐ Converges; $\frac{4}{7}$
- ☐ Diverges

17. What is the volume of the solid generated by rotating the region enclosed by the curves (4 Points)

$y = 2x$, $y = 2x^2$ for $0 \leq x \leq 1$ about the line $y = 4$.

- ☐ $\pi \int_0^2 \left[\left(4 - \frac{y}{2}\right)^2 - \left(4 - \sqrt{\frac{y}{2}}\right)^2 \right] dy$
- ☐ $\pi \int_0^2 \left[\left(4 - \sqrt{\frac{y}{2}}\right)^2 - \left(4 - \frac{y}{2}\right)^2 \right] dy$
- ☐ $\pi \int_0^1 [(4 - 2x)^2 - (4 - 2x^2)^2] dx$
- ☒ $\pi \int_0^1 [(4 - 2x^2)^2 - (4 - 2x)^2] dx$ ✓

18. (4 Points)

Which of the following is TRUE about the sequence $\left\{ \frac{n^2 - 1}{n} \right\}$

- ☐ It converges and increasing.
- ☐ It converges and decreasing.
- ☒ It diverges and increasing. ✓
- ☐ It diverges and decreasing.

19. (4 Points)

$\int \frac{\cos^3 x}{\sqrt{1 + \sin x}} dx$ is equal to

- ☒ $\frac{4}{3} (1 + \sin x)^{\frac{3}{2}} - \frac{2}{5} (1 + \sin x)^{\frac{5}{2}} + C$ ✓
- ☐ $(1 + \sin x)^{\frac{1}{2}} - \frac{1}{5} (1 + \sin x)^{\frac{3}{2}} + C$
- ☐ $\frac{1}{3} (1 + \sin x)^{\frac{3}{2}} - \frac{2}{5} (1 + \sin x)^{\frac{5}{2}} + C$
- ☐ $\frac{4}{3} (1 + \sin x)^{\frac{5}{2}} - \frac{2}{5} (1 + \sin x)^{\frac{3}{2}} + C$

20. (4 Points)

Which of the following is TRUE about the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sqrt{n+3}}{n}$.

- ☒ The series is conditionally convergent. ✓
- ☐ The series is absolutely convergent.
- ☐ The series is divergent.
- ☐ None of the shown answers.

21. (4 Points)

One of the following statements is TRUE

- ☐ If a sequence is bounded, then it is convergent.
- ☐ If a sequence is decreasing, then it is convergent.
- ☐ If $\lim_{n \rightarrow \infty} a_n = 0$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.
- ☒ If the series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$. ✓

22. Find a formula for the nth-term of the sequence (4 Points)

$n^3 - \frac{n^9}{3!} + \frac{n^{15}}{5!} - \dots$

- ☐ $a_n = (-1)^{n+1} \frac{(n^3)^{2n-1}}{(n-1)!}$, $n \geq 1$
- ☐ $a_n = (-1)^{n+1} \frac{(n^3)^{2n}}{(2n-1)!}$, $n \geq 1$
- ☐ $a_n = (-1)^n \frac{(n^3)^{2n}}{(2n+1)!}$, $n \geq 0$
- ☒ $a_n = (-1)^n \frac{(n^3)^{2n+1}}{(2n+1)!}$, $n \geq 0$ ✓

23. (4 Points)

The binomial coefficient $\binom{-2}{6}$ is equal to

- ☐ -7
- ☒ 7 ✓
- ☐ $\frac{7}{6!}$
- ☐ $-\frac{7}{6!}$

24. (4 Points)

Using $t = \tan\left(\frac{x}{2}\right)$ the integral $\int \frac{\sin x}{1 + \sin x - \cos x} dx$ leads to

- ☐ $\int \frac{2}{(t+1)^2} dt$
- ☐ $\int \frac{2}{(t^2+1)^2} dt$
- ☐ $\int \frac{2}{(t-1)(t^2+1)} dt$
- ☒ $\int \frac{2}{(t+1)(t^2+1)} dt$ ✓

25. (4 Points)

Which of the following is TRUE

- ☒ $\int \tan^n x dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x dx$, $n > 1$ ✓
- ☐ $\int \tan^n x dx = \frac{\tan^{n-1} x}{n-1} + \frac{n-2}{n-1} \int \tan^{n-2} x dx$, $n > 1$
- ☐ $\int \tan^n x dx = \frac{\tan^{n-1} x}{n} + \frac{n-1}{n} \int \tan^{n-2} x dx$, $n > 1$
- ☐ $\int \tan^n x dx = \tan^{n-1} x + n \int \tan^{n-2} x dx$, $n > 1$